AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

Course Code: FIN240

Course Title: Microfinance Theory and Practice

Final Exam VA

Spring 24-25
Date: 3-June-2025

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Number of exam pages: 5 (including this cover page)	Time Allowed: 3 hours

Instructions:

1. Take FIVE (5) minutes to read all instructions carefully before attempting to write.
2. Mobile phones must be in "SWITCHED OFF" mode throughout the examination.
3. This is a CLOSED book examination. NO documents of any kind are allowed.
4. You MUST answer the questions using only the answer booklets provided
5. You MUST return the examination paper together with your answer booklets.
6. Do NOT write your answers in pencil except graphs/sketches
7. Students are NOT allowed to use any type of dictionary or mobile phones during the examination
8. Students are allowed to use non-programmable calculators only (if needed).
9. The penalty for an ACADEMIC MISCONDUCT in this examination is an F Grade.

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Part A: Solve BOTH Problems (60 Points)

Problem 1: (15 marks)

Given:
the following portfolio and aging of arrears report of ABC microfinance.

AGING OF ARREARS - 2021					
Days in arrears	Number of loans	Outstanding balance	Amount in arrear	Reserve rate	Reserve amount
Current loans	4,800	720,000	0		
< 30 days	2,250	330,000	34,500	10%	33,000
31-60 days	1,500	180,000	19,851	25%	45,000
61-90 days	750	75,000	15,000	50%	37,500
91-120 days	225	31,500	9,000	75%	23,625
> 120 days	75	13,500	6,000	100%	13,500
TOTAL	**9,600**	**1,350,000**	**84,351**		**152,625**

PORTFOLIO REPORT		
	Dec-21	Dec-20
Total value of loans disbursed	4,800,000	1,500,000
Total number of loans disbursed	22,500	9,750
Number of loans outstanding	9,600	3,750
Value of loans outstanding	1,350,000	675,000
Total of payments in arrears	84,351	63,225
Outstanding balance of loans in arrears	630,000	270,000
Value of loans written off during period	45,000	22,000
Average loan disbursement size	213	154
Average loan balance size	141	180

Required:
Analyze the quality of this portfolio by calculating:

- the arrears rate for loans with payments more than 60 days overdue (5 marks)
- the portfolio at risk (PAR) for loans with payments more than 90 days overdue (5 marks)
- the Loan Loss Rate. (5 marks)

Problem 2: (45 marks)

Starting from the balance sheet, the income statement below and the information about inflation, subsidies and donations, you are required to make the necessary adjustments and evaluate the finan viability of the XYZ MFI under analysis (OSS and FSS).

Balance Sheet		
	Dec. 2021	Dec. 2020
ASSETS		

Income Statement		
	Dec. 2021	Dec. 2020
OPERATING INCOME		

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Cash	15,500	15,000
Investments	40,000	50,000
Total loan portfolio	425,000	300,000
(Loan loss reserve)	(32,500)	(25,000)
Total net loans outstanding	392,500	275,000
Fixed assets	17,250	12,250
(Accumulated depreciation)	(5,000)	(4,000)
Net fixed assets	12,250	8,250
TOTAL ASSETS	460,250	348,250
LIABILITIES		
Forced savings	60,000	50,000
Commercial loans	5,000	5,000
Subsidized loans	17,500	17,500
TOTAL LIABILITIES	82,500	72,500
EQUITY		
Donated equity cumulated	225,000	200,000
Donated equity current	0	25,000
Cumulated earnings/losses	50,750	(25,000)
Current earnings/losses	102,000	75,750
TOTAL EQUITY	377,750	275,750
TOTAL LIABILITIES + EQUITY	460,250	348,250

Interests and fees from loans	204,300	157,350
Income from investments	12,500	10,000
TOTAL OPERATING INCOME	216,800	167,350
OPERATING EXPENSES		
Interest and fee expenses	1,100	1,000
Loan loss provision expense	17,500	5,000
Salaries	77,500	62,500
Other administrative expenses	17,500	22,500
Depreciation	1,200	600
TOTAL OPERATING EXPENSES	114,800	91,600
NET OPERATING PROFIT / LOSS	102,000	75,750
NON OPERATING INCOME		
Cash donations	0	25,000
Other non-operating income	0	0
TOTAL NON OPERATING INCOME	0	25,000
Non-operating expenses	0	0
TOTAL PROFIT / LOSS	102,000	100,750

Inflation adjustments

The current rate of inflation is 5%.

Subsidies adjustments

The market rate for commercial loans is 10%. The actual interest expense for subsidized loans is US$500

In-kind donations adjustments

The MFI received free equipments and technical assistance from an international donor, for a total market value of US$100,000.

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Part B: Choose 2 out of 3 Questions (40 Points)

Question 3:

It has been argued that FinTechs help innovate the business model of financial services firms and its effect on the microfinance business model is fourfold, one of them is Revenue Structure. You are required to discuss how FinTech affects **customer retention** and **customer acquisition. (20 marks)**

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Question 4:

MFIs like similar organizations use product pricing strategies to set a fair price for their products, you are required to define product pricing and briefly discuss the three primary methods used by MFIs. **(20 marks)**

Question 5:

Insurance products should be largely designed based on demand research. What is it that people want coverage for? Microinsurance products can be broadly classified into five primary categories. Discuss Life and Agricultural insurance. Support your answer with relevant examples. **(20 marks)**

***************End of questions****************

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FIN240 Formula Sheet

- APR = (AE + LL + CF + K –II) / (1-LL)
- AE = (salaries + rent + depreciation + other administrative expenses) / Average outstanding portfolio
- LL = ((Loan loss reserve year(n-1) + Loan loss provision year(n))- Loan loss reserve year(n) / average outstanding portfolio
- Average equity for cost of funds calculation = Average equity – average net fixed assets
- Forced savings = Average forced savings x rate on saving services
- subsidized loan= Average subsidized loans x rate on commercial loans
- Commercial loan = Average commercial loans x rate on commercial loans
- Equity= Average equity for cost of funds calculation x inflation rate
- CF = (Forced savings + subsidized loan + Commercial loans + Equity) / Average outstanding portfolio. note: the numbers in the numerator are used after applying the market rates
- II = Income from other than loan portfolio / Average outstanding loans
- Arrears Rate = Total loan installments past due / Gross portfolio outstanding
- Portfolio at Risk (PAR) = Total outstanding loan balance on delinquent loans / Gross portfolio outstanding

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- Loan Loss rate = total amount written off (in the period) / Average outstanding portfolio (in the period)
- INFLATION ADJUSTMENT = (Average equity – Average net fixed assets) x Inflation rate
- COST OF FUNDS ADJUSTMENT = Average subsidized funds x (commercial interest rate – interest rate paid) OR *(Average subsidized funds x commercial rate) - interest rate paid*
- OSS = operating income / total operating expenses
- FSS = Adjusted operating income / adjusted operating expenses

Ratio	Formula
Portfolio to assets	Gross loan portfolio/assets
Operating efficiency ratio	Operating expense/revenues
Cost per active client	Operating expense/average number of active clients
Borrowers per loan officer	Number of active borrowers/ number of loan officers
Active clients per staff member	Number of active clients/total number of personnel
Portfolio yield	Cash received from interest, fees, and commissions on loan portfolio/ average gross loan portfolio
Return on assets (ROA)	Net income after taxes and before donations/average assets
Return on equity (ROE)	Net income after taxes and before donations/average equity
Nonperforming loans as of 30 days past due (NPL30)	(Nonperforming loans> 30 days + value of renegotiated loans)/gross loan portfolio
Renegotiated loans ratio	Value of renegotiated loans/gross loan portfolio
Write-off ratio	Value of loans written off/average gross loan portfolio
Debt to equity (leverage)	Liabilities/equity
Equity to assets ratio	Total equity/total assets
Liquid ratio (quick ratio or acid-test ratio)	(Cash + trade investments)/ (demand deposits + short-term time deposits +short-term borrowings + interest payable on funding liabilities +accounts payable + other short-term liabilities)
Liquid assets to total assets	Cash/total assets
Loans to deposits ratio	Gross loan portfolio/deposits

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